(Another) Port Opening as an Industrial Policy

April 29, 2025

- How should countries promote certain industries?
- How can export promotion policies help?
- Does all this matter for countries' growth?

Export Promotion – Port Openings

Region/Country	**Port Examples**
United States	Houston, LA, Seattle
EU	Rotterdam, Hamburg
Brazil	Santos, Rio de Janeiro, Paranaguá
China	Ningbo, Tianjin, Shenzhen
India	Mumbai, Chennai
Southeast Asia	Singapore, Port Klang
Ghana and Côte d'Ivoire	Tema
China (BRI projects)	Chancay (Peru)

Port openings – Reduce export costs for nearby regions

$\Rightarrow$ Export subsidy for nearby regions

- Regions: More entrants, including multi-regional firms
- Firms: Better technology/More employment for exporting

$\Rightarrow$ More competition for the regions using established ports

- Regions: Exit of less productive firms
- Firms: Innovate to escape competition

Port openings – Challenges

- Endogenous
- Complicated general equilibrium effects
- Limited firm-level data

Port openings – Challenges

- Endogenous
  $\Rightarrow$ Sudden alternative port opening due to a natural disaster

- Complicated general equilibrium effects
  $\Rightarrow$ "Port opening" only for a single good

- Limited firm-level data
  $\Rightarrow$ Census of firms across time

$\Rightarrow$ Silk industry in pre-WW2 Japan

Industry Policy and Export Promotion

- Industrial policy for a developing region

  Industrial Policy (Barwick et al., 2024), Place-based policy (Fajgelbaum & Gaubert, 2025)

- Benefits of subnational export promotion policy

  Technology upgrading (Verhoogen, 2023), Innovation & Allocation (Melitz & Redding, 2021),

  Multinationals (Antras & Yeaple, 2014), Intranational trade cost (Fajgelbaum & Cosar, 2016)

- Empirical evidence on port opening for broader regions

  Port investment (Brancaccio et al., 2024),

  Port city growth (Ducruet et al., 2022) (Brooks et al., 2021)

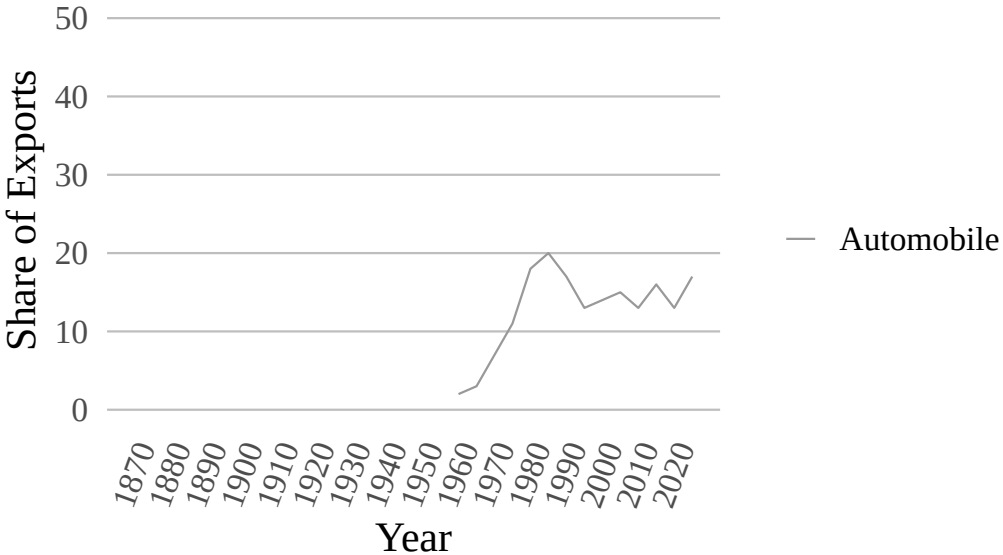
Outline

Setting and Descriptive

Framework and Counterfactual

Setting and Descriptive

Japan: Export-led Industrialization



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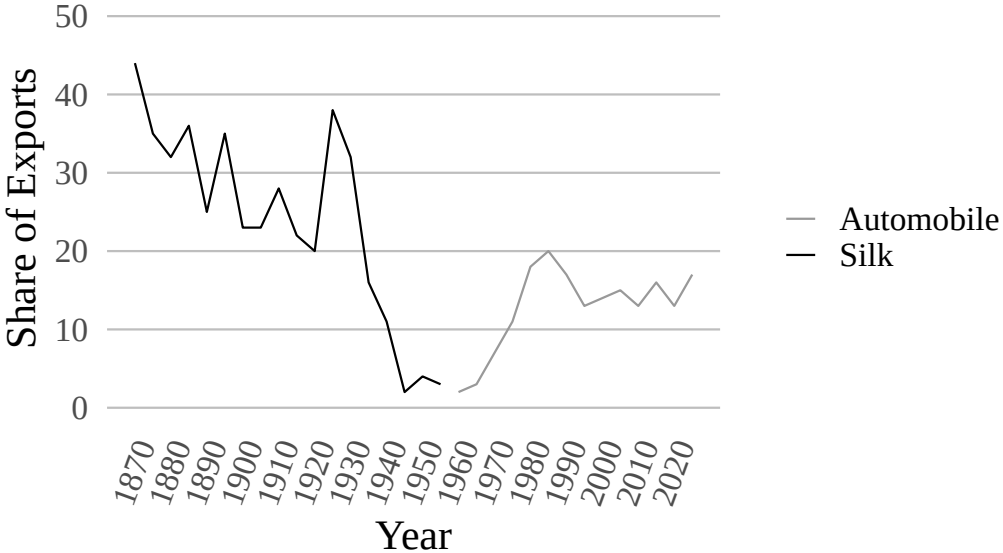

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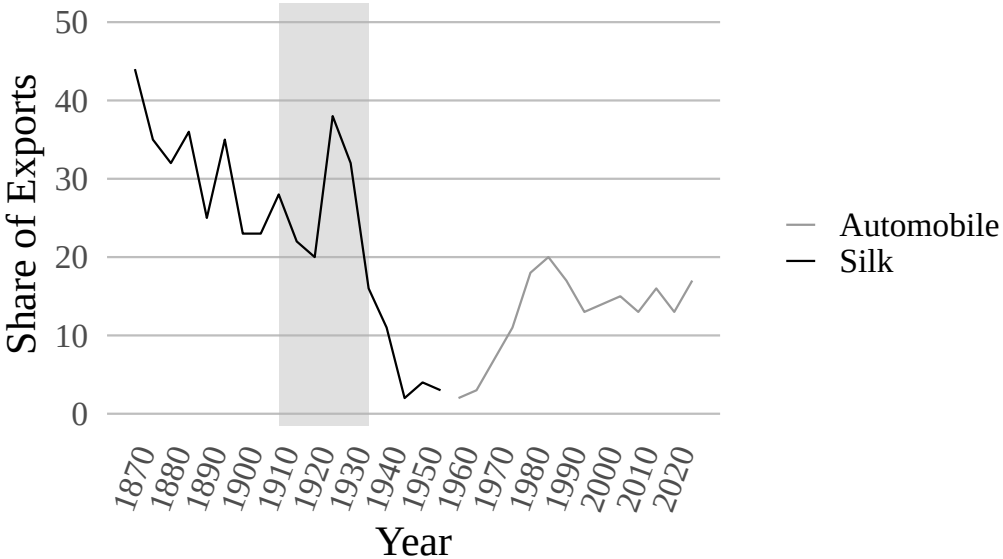
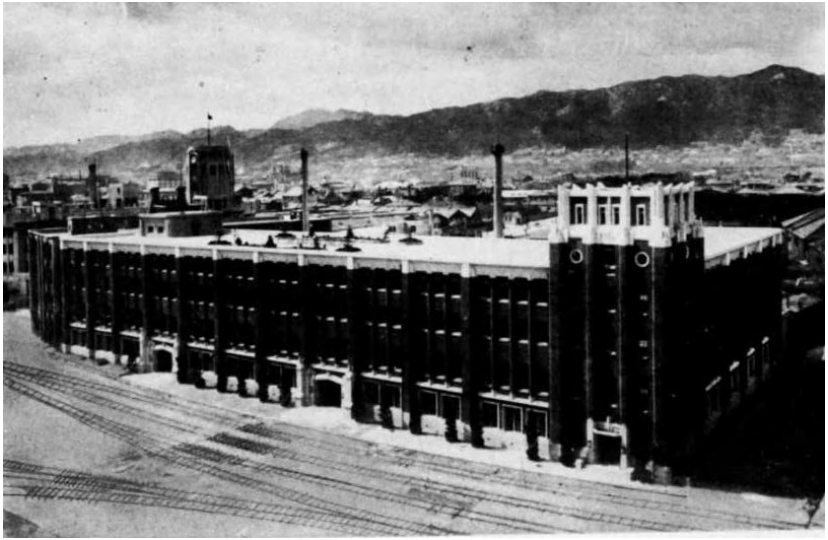


Export-oriented industry: The Japan to U.S. Silk Road

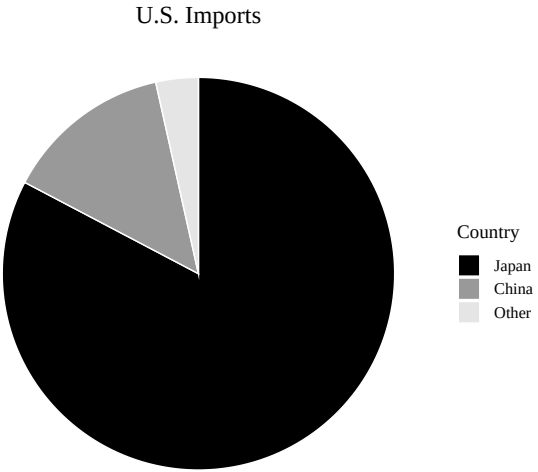
Cocoon (Japanese Farmers) → Raw Silk (Japanese Factories) → Silk Products (U.S. Firms)

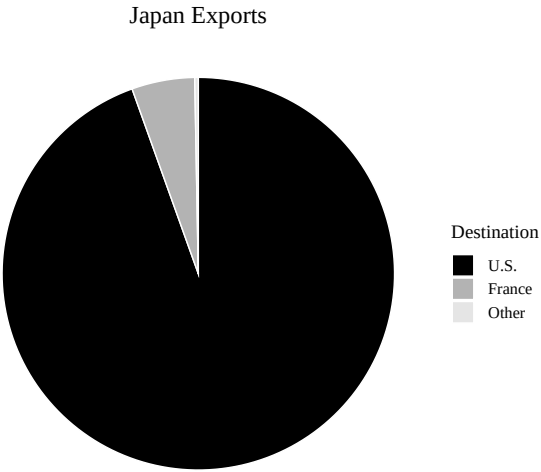
Reeling Machines
Female Workers

High quality exports: Inspection office required

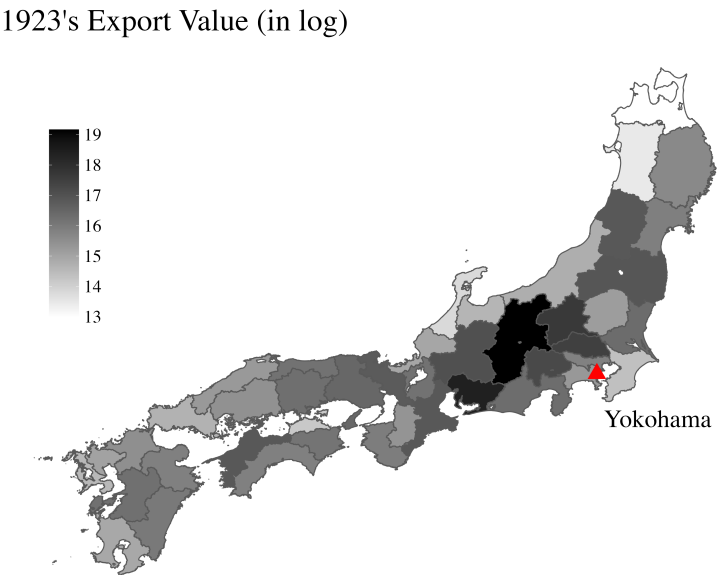


The Japan to U.S. Silk Road





Natural experiment: Earthquake in 1923



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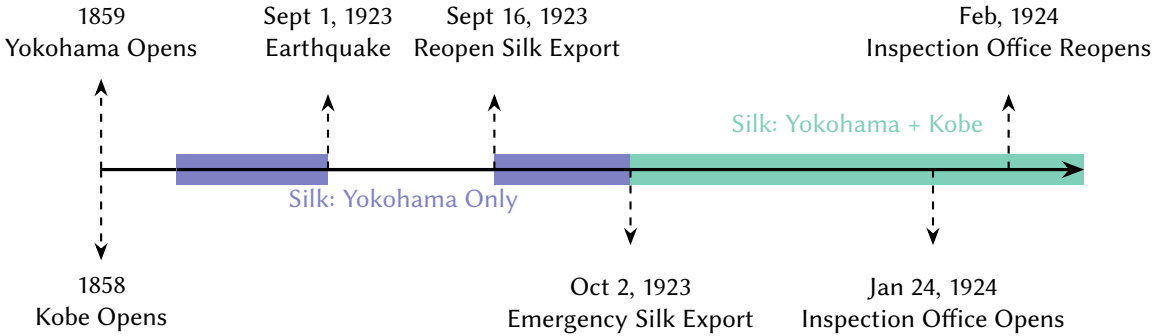
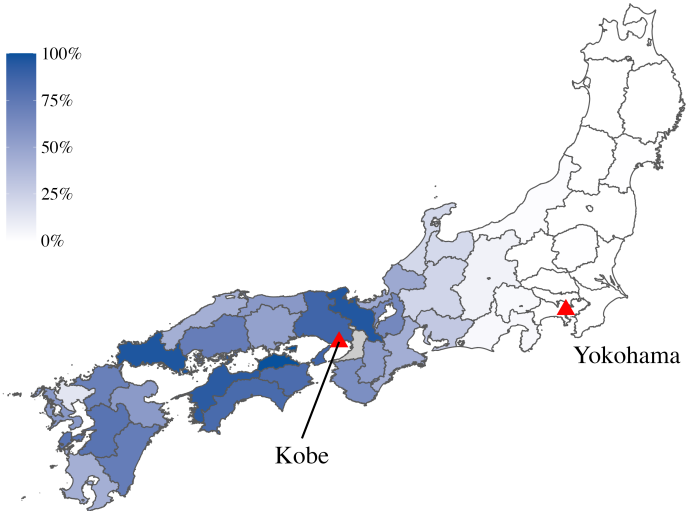


Natural experiment: Earthquake in 1923

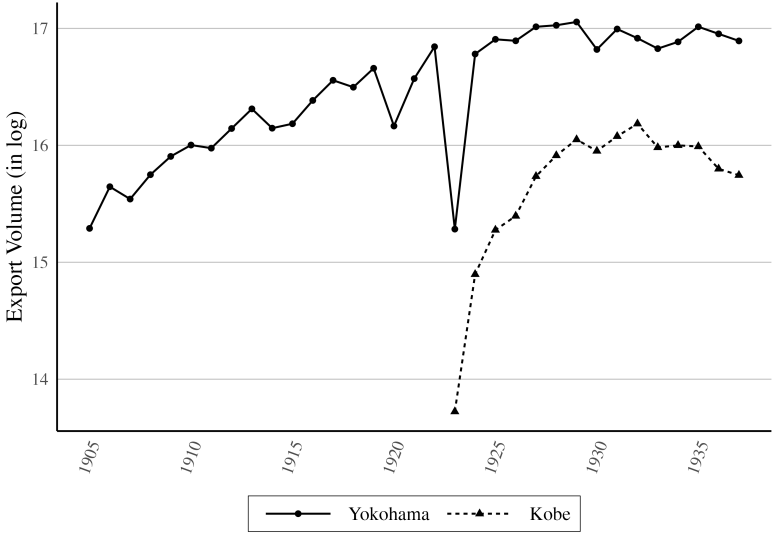
Share of Silk Shipment to Kobe Port in 1931



Natural experiment: An exogenous port opening

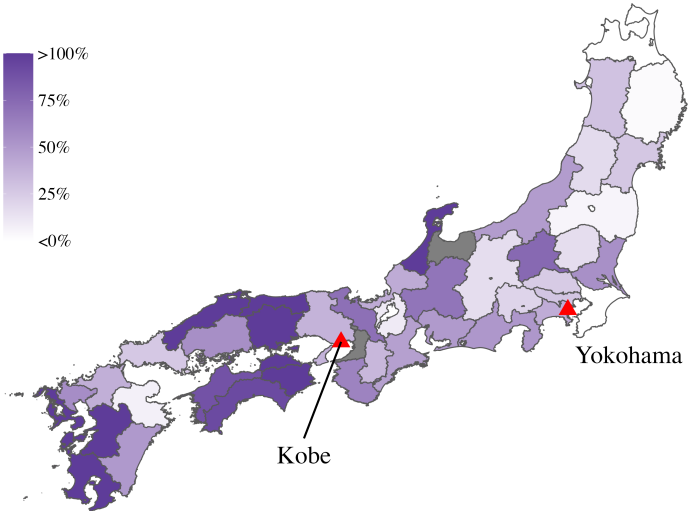
- -1923: Exporting firms near the only exporting port (East Japan)
- 1923-: Exporting opportunity to a less developed region (West Japan)
- How did different regions and the silk industry respond?

Single to Dual-Port Equilibrium



Export Growth in the West

Silk Export Growth, Machine (1924-1930)



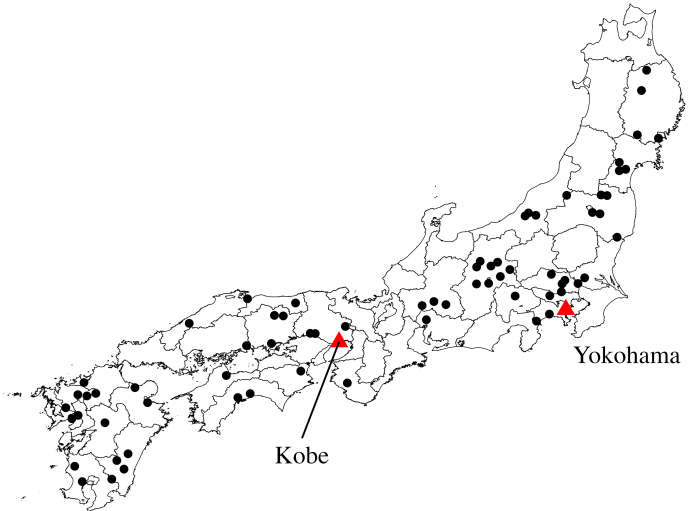
Treated silk regions grow (suggestive)

$$y_{it} = \beta \underbrace{\Delta Port\ Access_i}_{\min\{\log(dist_{i,Yokohama} - dist_{i,Kobe}),\ 0\}} \times After_t + \alpha_i + \lambda_t + \varepsilon_{it}$$

	N of Factories (1)	Labor (2)	Capital (3)	Silk Output (4)
Δ Port Access $\times$ After	0.400	1.131**	1.143**	1.548***
	(0.502)	(0.473)	(0.490)	(0.562)
Prefecture FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Observations	129	129	129	129
Adjusted R^2	0.85	0.83	0.85	0.83

Multi-regional firms: Importance?

Katakura Plant Locations



To understand industry dynamics and regional growth

Dig into the census on silk-reeling firms ($\approx$ biennial)

- Who
  - Multi-regional firms, Existing firms, New entrants
- Location
  - Close to low input cost or better exporting locations
- Time
  - After the port opening
  - Macroeconomic shocks (post WW1, the Great Depression)
- Firm decision
  - Upgrade their technology/capacity (types/number of machines)
  - Enter/Exit/Branch out
- Driving factors
  - Input costs (cocoon cost, wage)

Framework and Counterfactual

The effect of the port opening using Ericson & Pakes 1995

- Consumers in the U.S.
  - Buy cheaper/better silk
- Firms in West Japan
  - Expand by buying (better) machines
  - Enter (start-up or multi-regional from the East)
- Firms in East Japan
  - Buy better machines to escape competition
  - Exit

States

- Incumbent firm $i \in \mathcal{I}$ has state $\omega_{it} = (r_i, a_i, \psi_{ikt}, m_{ikt})$
  - Region $r_i \in \{W, E\}$
  - Permanent TFP a_i
  - Machine type (quality) ψ_{ikt}
  - Machine quantity m_{ikt}

- Aggregate state $\omega_t = (\omega_{1t}, \ldots.\omega_{|\mathcal{I}|t}, \tau_{Wt}, \tau_{Et}, w_{Wt}, w_{Et}, d_t)$
  - Trade cost τ_{rt}
  - Input cost w_{rt}
  - Demand shock d_t

Actions

- Potential Entrants to West Japan
  - Start-up draws $a_i \sim G_W$ and pays f_W
  - Multi-regional draws $a_i \sim G_E$ and pays $f_W + f_{MR}$
- Incumbents
  - Exit
  - Buy the same machine Δm_{ik} with cost $c_k(\Delta m_{ik})$
  - Upgrading machines $m_{ik'}$ with cost $c_{k'}(m_{ik'})$

State Transition

- Productivity distribution
  - More firms enter in the West
  - Some are multi-regionals pushing up the West's distribution
  - East also has better productivity due to exits
- Machine quality and quantity distribution
  - East and West invest in new machines
- Demand
  - Numerous depressions dampen investments in the 1920s

Back to the main questions

- How should countries promote certain industries?
  Silk industry growth for different subsidies
  - Entry subsidy ... f_W
  - Multi-regional subsidy ... f_{MR}
  - Input subsidy ... w_r
  - Technology adoption ... $c_k(\cdot)$

- How can export promotion policies help?
  Silk industry growth when trade vs domestic policies
  - Export subsidy ... τ_W
  - Bigger market ... d_t
  - Timing of subsidy ... difference in G_W, G_E

Things to do

- Does all this matter for countries' growth?

  Changes in regional employment/wages and long-term growth

- Framework
  - Add Japanese workers to evaluate regional growth
  - Multiple regions within East and West?
  - Domestic silk consumption?
  - Make computation feasible (distributional assumption on productivity?)

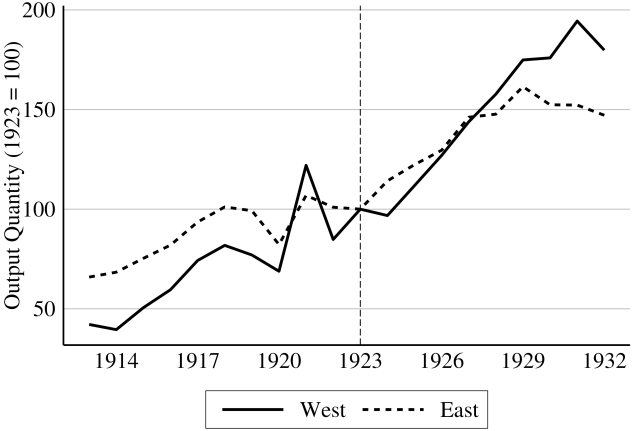
- Data: Digitize

- Empirics: Use exogenous timing of the port opening

Thank you!

Appendix

Treatment Effect?



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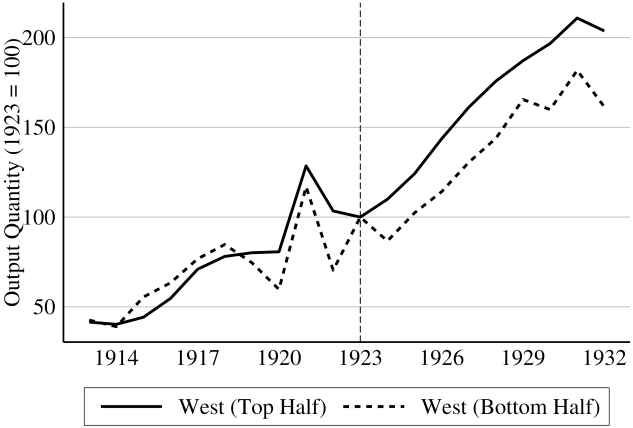
Treatment Effect (Within the West)



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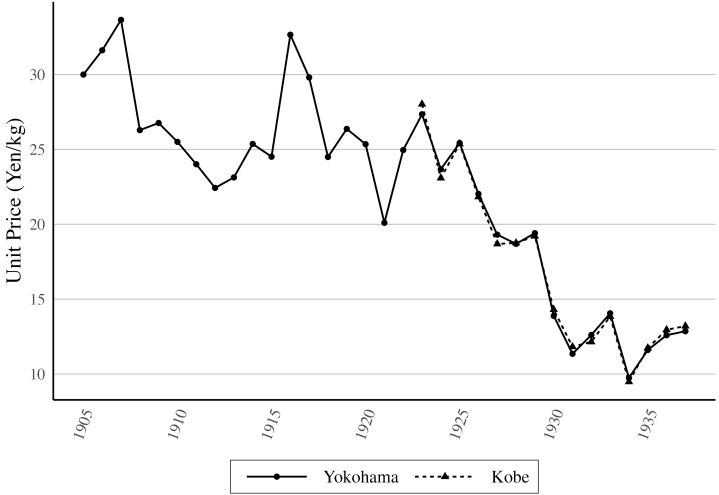
Single to Dual-Port Equilibrium – Price



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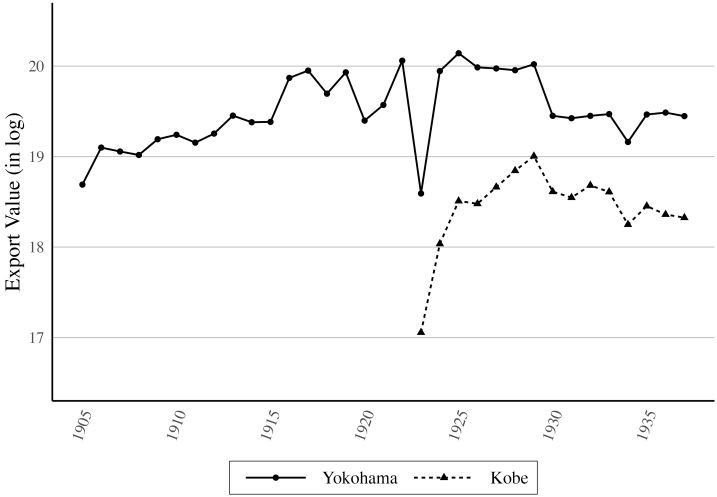
Single to Dual-Port Equilibrium – Value